Qiyan Jin

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Mechanical Engineering graduate and NTU MSc student focused on robot manipulation learning, with experience in π0.5 training and evaluation, Unitree G1 teleoperation and cross-simulator testing of dexterous hands.

Education

Nanyang Technological University, Singapore — **Aug 2026 - May 2027**

School of Electrical and Electronic Engineering (EEE)

Master of Science (Computer Control & Automation) | Full-time; expected completion May 2027

Shanghai Jiao Tong University — **Sep 2022 - Aug 2026**

University of Michigan-Shanghai Jiao Tong University Joint Institute

Bachelor of Science in Engineering, Mechanical Engineering | Graduated | GPA: 3.2 / 4.0

Experience

Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR) — **May 2026 - Sep 2026**

Intern - Dexterous-Hand Simulation Benchmark (DexArena)

> **Project:** DexArena is a simulation benchmark for dexterous manipulation across multiple hand models, with everyday-object tasks for policy training and evaluation.

- Developed and adapted dexterous-hand tasks in ManiSkill, refining observations, rewards and success checks for different hand models, and checking task behaviour during training and evaluation.
- Contributed to π0.5 data adaptation, supervised fine-tuning and closed-loop evaluation. Checked joint ordering, control interfaces and visual inputs, and debugged configuration and interface issues.
- Helped write the DexArena manuscript's methods and experimental setup, checking descriptions against training settings and evaluation records.

Selected Projects

Unitree G1 Whole-Body VLA Simulation — **May 2026 - Aug 2026**

Undergraduate capstone | 5-person team

> **Project:** The team explored humanoid whole-body manipulation in MuJoCo, using teleoperated demonstrations to train GR00T N1.7 policies and SONIC WBC for whole-body control.

- Primarily handled PICO 4 / VR_3PT whole-body teleoperation. Helped calibrate the headset, controllers and simulated robot, and collected demonstrations of walking, grasping and bimanual handover.
- Reviewed task labels, episode boundaries and action completeness in long-horizon demonstrations; identified control anomalies and recollected low-quality trajectories for the team's training dataset.
- Reproduced control degradation when video recording was enabled, contributed to collection-software and workflow changes, and rechecked teleoperation with recording enabled.

Sharpa Wave Model Checks and Cross-Simulator Evaluation — **Aug 2026 - Sep 2026**

Independent project

> **Project:** Checked publicly available Sharpa Wave hand models and compared motion and simplified contact behaviour in MuJoCo and kit-less OVPhysX.

- Built model checks for joints, motion limits, mesh paths and control settings. Tested loading and motion, and found mesh-path issues in some dual-hand models.
- Compared joint motion, fingertip trajectories and simplified contact responses under the same scenes and control inputs. Fingertip trajectory differences were repeatable and exceeded the accepted tolerance.
- Traced contact readout changes to how simulation steps were called. Adjusted the calls to stabilise contact detection without changing motion trajectories; contact timing and blocked travel still differed between simulators.

Technical Skills

Programming & Learning	Python, MATLAB, PyTorch; LeRobot, π0.5 supervised fine-tuning
Simulation & Evaluation	ManiSkill, MuJoCo, kit-less OVPhysX; task development, policy evaluation
Teleoperation & Data	Teleoperation calibration, demonstration collection and quality review
Design & Fabrication	SolidWorks / CAD, 3D printing, laser cutting, prototype assembly and testing

Additional Experience

Ronovo Surgical | Project Assistant Intern
Feb 2025 - Apr 2025

- Supported medical-robotics project scheduling and R&D milestone tracking, coordinating updates across hardware, software testing and supply chain teams.
- Organised technical documents and bills of materials, followed up on component procurement, and assisted with prototype assembly and basic testing.

Additional Engineering Projects

Dual-Arm Dental Navigation Simulation
Junior Fall

ME4670J Introduction to Robotics

- Derived a six-frame coordinate-transform chain between surgical and vision arms for an occluded target. Used Python and MATLAB to test inverse kinematics, configuration space and RRT* path planning.

Cylindrical Pier Inspection Climbing Robot
Junior Summer

ME3500J Design and Manufacturing II

- Managed scheduling and procurement; iterated a three-drive-unit chassis and spring-pin/one-way-slot self-locking linkage in CAD. Fabricated PLA and acrylic parts, then assembled and tested the prototype.

Automatic Card Dealer for Multiple Players
Sophomore Summer

ME2500J Design and Manufacturing I

- Designed the dealing mechanism and top-mounted electronics to rotate together, reducing cable winding. Adjusted the card-release lead for rotational speed to deal continuously to different positions.

Mobile Robot for Area Light-Intensity Mapping
Freshman Summer

ENGR1000J Introduction to Engineering

- Modelled the chassis in SolidWorks, helped debug the light sensor, and used Python to turn collected readings into an area light-intensity map.

Teaching and Campus Experience

Teaching Assistant | ME3950J Experiment (1)
Senior Fall

- Handled the Technical Communication component's formatting checks, helped grade assignments, and marked content that did not follow course requirements.

Teaching Assistant | ME2500J Design and Manufacturing I
Junior Spring & Summer

- Ran lab sessions and taught CAD across two course offerings. Guided mechanical design, component selection, basic machining and presentation preparation; helped with grading, grade entry and Canvas materials.

Academic Administration Assistant
Mar 2024 - Aug 2024

Undergraduate Education Office, UM-SJTU Joint Institute

- Supported course scheduling, TA coordination, course evaluation, examinations and grade administration, as well as course-selection guidance and degree audits.

Honours and Competitions

- First Prize, Shanghai Korean-Chinese University Student Scholarship | Mar 2024
- First Prize, ME2500J Game Day
- Participation Award, SJTU RoboMaster Campus Competition | Dec 2022
- Third Prize, SJTU "Liming Cup" Mechanical Creativity Competition | Apr 2023
- Outstanding Design Award, Summer Program Expo | 2022-2023 Academic Year